

Heart Disease Analysis

Project Design Phase-II: Solution Requirements (Functional & Non-functional)

Project Meta-Information

Date	10 July 2026
Team ID	—
Maximum Marks	4 Marks

Functional Requirements

FR No.	Functional Requirement	Sub Requirement (Story/Sub-Task)
FR-1	Dashboard Visualization	Interactive Tableau dashboard with KPIs: Total Patients, Heart Disease Cases, Average BMI.
FR-2	Heart Disease Distribution	Analyze heart disease prevalence by Age Category and Gender.
FR-3	BMI & Lifestyle Analysis	Compare BMI with Smoking, Alcohol Drinking and Physical Activity.
FR-4	Medical Condition Analysis	Analyze Diabetes, Stroke, Kidney Disease, Asthma and Skin Cancer with Heart Disease.
FR-5	General Health Analysis	Analyze General Health, Physical Health, Mental Health and Sleep Time.
FR-6	Demographic Analysis	Compare disease prevalence across Age, Gender and Race.
FR-7	Interactive Filters	Filter by Age, Gender, BMI, Smoking, Diabetes, Physical Activity and General Health.
FR-8	Tableau Story	Present guided healthcare insights and heart disease trends.
FR-9	Flask Website	Embed Tableau Dashboard and Story in Flask web application.
FR-10	Healthcare Reporting	Generate reports for doctors and hospital administrators.

Non-Functional Requirements

NFR No.	Requirement Type	Description
NFR-1	Usability	Easy navigation with intuitive healthcare dashboards.

NFR No.	Requirement Type	Description
NFR-2	Security	Protect patient information and securely display analytics.
NFR-3	Reliability	Provide accurate results using the Heart_new2 dataset.
NFR-4	Performance	Fast dashboard loading and responsive filters.
NFR-5	Availability	Dashboard and Flask application should be continuously accessible.
NFR-6	Scalability	Support larger datasets and future predictive analytics.